

A PROJECT REPORT

ON

“CampusHub”

Carried Out at



**CENTRE FOR DEVELOPMENT OF ADVANCED
COMPUTING Anusandhan Bhavan, sector 62, Noida**

UNDER THE SUPERVISION OF

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PG DIPLOMA IN ADVANCED COMPUTING, NOIDA

CANDIDATE'S DECLARATION

We hereby certify that the work being presented in the report entitled “**CampusHub**”, in partial fulfillment of the requirements for the award of PG Diploma Certificate and submitted in the department of PG-DAC of the C-DAC Noida, is an authentic record of our work under the supervision of **Mrs.Anu Mehra** , C-DAC Noida.

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ACKNOWLEDGMENT

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CERTIFICATE

This is to certify that the work titled “**CampusHub**” is carried out by Shikhar Sharma (250820520005), the Bonafide Students of Diploma in Advanced Computing Anusandhan Bhavan, Noida. The Course End Project work is carried out under my direct supervision and completed.

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Abstract

In the modern educational landscape, the management of academic and administrative data is often hindered by fragmented manual processes, leading to data redundancy, inefficiency, and lack of real-time accessibility. This project, **CampusHub**, presents a comprehensive, web-based University Management System designed to digitize and streamline the core operational workflows of an educational institution.

Built on a robust **3-Tier Architecture**, the system utilizes **React.js** for a responsive, user-friendly frontend and **Java Spring Boot** for a secure, scalable backend. Data persistence is managed via **PostgreSQL**, an Object-Relational Database selected for its strict adherence to ACID properties and ability to handle complex relational data.

Chapter 1.0

Introduction

In the traditional operational framework of many educational institutions, administrative and academic processes are often managed through disparate, manual systems. This reliance on physical paperwork and isolated digital records creates significant challenges, including data redundancy, slow information retrieval, and a lack of coordination between departments such as the Registrar, Faculty, and Examination cells. As universities grow in size and complexity, the need for a unified digital solution becomes critical to ensure efficiency, accountability, and seamless communication across the campus ecosystem.

CampusHub is designed as a robust, web-based University Management System to address these challenges by centralizing essential academic workflows into a single, integrated platform. Built upon a secure 3-tier architecture, the system leverages the responsiveness of **React.js** for the frontend, the stability of **Java Spring Boot** for the backend, and the data integrity of **PostgreSQL** for storage. This modern technology stack ensures that the application is not only scalable and fast but also capable of maintaining the strict consistency required for handling sensitive student and employee records.

The core philosophy of CampusHub is **Role-Based Access Control (RBAC)**, which ensures that every stakeholder—whether a Student, HOD, Faculty member, or Administrator—interacts with a tailored interface relevant to their specific duties. By digitizing key processes such as student registration, profile management, and examination scheduling, CampusHub eliminates the inefficiencies of manual data entry. The result is a transparent, paperless environment where data flows seamlessly between departments, allowing the administration to focus less on clerical tasks and more on academic excellence.

Problem Statement:

In the current operational landscape of many higher educational institutions, the management of academic and administrative processes is severely hampered by a reliance on manual, paper-based workflows and fragmented legacy systems. This decentralized approach creates significant data silos where critical information—such as student profiles, academic history, and employee records—is duplicated across various departments like the Registrar's office, the Examination Cell, and Faculty departments. Such redundancy not only consumes excessive physical storage and administrative man-hours but also leads to inevitable data inconsistencies, where a change in a student's record in one department is not reflected in others, causing confusion and operational delays.

Furthermore, the lack of a centralized, automated system results in critical gaps in data integrity and validation. Without robust digital constraints, institutions frequently struggle with "dirty data," such as incomplete student profiles, invalid contact details, or mismatched academic records, which directly impact decision-making and reporting. The absence of real-time validation means that errors are often detected only when they cause a critical failure, such as a student being unable to register for exams due to missing semester information. Additionally, the security of sensitive personal data is often compromised in manual systems, where physical files are vulnerable to loss, damage, or unauthorized access.

Finally, the existing communication channels between the administration and the student body are often inefficient and unidirectional. Vital information, such as examination schedules, syllabus updates, or urgent notices, is typically disseminated via physical notice boards, leading to information latency and limited accessibility for students off-campus. This disconnect forces students to physically visit administrative blocks for routine queries, resulting in overcrowding and wasted academic time. Consequently, there is an urgent need for an integrated, web-based solution that digitizes these workflows, enforces strict data validation, and provides a secure, unified interface for all university stakeholders.

Scope and Objective

Objective:

The primary goal of CampusHub is to transition the university from a manual, paper-dependent operational model to a fully digital, automated ecosystem.

- **Centralization of Data:** To consolidate scattered records (from Registrar, Departments, and Exam Cell) into a single, unified database to eliminate redundancy and data silos.
- **Automation of Workflows:** To replace manual registration and scheduling processes with automated digital workflows, significantly reducing administrative processing time.
- **Data Integrity & Validation:** To implement strict validation protocols (both client-side and server-side) that ensure only accurate, consistent, and complete data enters the system.
- **Role-Based Security**
- **Real-Time Accessibility**

Scope:

A. User Roles & Modules

- **Administrator (Registrar):** Capability to manage master data (Courses, Branches, Subjects) and oversee the entire system.
- **Student Module:** Features for secure registration, profile management (personal and academic details), and viewing personalized exam schedules.
- **Employee Module:** Specialized interfaces for Faculty, HODs, and Exam Controllers to manage their profiles and perform role-specific duties.
- **Examination Cell:** Functionality to create, update, and publish exam schedules mapped to specific branches and semesters.

B. Functional Scope

- **Secure Authentication:** Implementation of a secure login system using JWT (JSON Web Tokens) and BCrypt password hashing for all user types.
- **Profile Management:** Comprehensive forms for capturing and updating detailed user information (Personal, Academic, Family) with robust validation constraints.
- **Master Data Management:** A dynamic database-driven approach to managing university curriculum structures (Departments, Courses) to avoid hardcoding.
- **Search & Retrieval:** Efficient logic to fetch and display data (e.g., Schedules) based on logical mapping (e.g., specific Semester and Branch).

Chapter 2.0

Software Requirement Specification

Functional requirements:

A. Authentication Module

- **Login**
- **Role-Based Access**
- **Change Password**

B. Student Module

- **Registration**
- **Duplicate Check**
- **View Profile**
- **Update Profile**
- **View Schedule**

C. Employee & Admin Module

- **Employee Registration**
- **Master Data Management**
- **Exam Scheduling**
- **Employee Profile Management**

Other non functional requirements:

- **Data Encryption**
- **Access Control**
- **Validation**
- **Response Time**
- **Database Latency**
- **Concurrency**
- **Uptime**
- **Data Consistency**
- **Database Scalability**
- **Modular Architecture**
- **Client Side**

Following are the software used

Software used	Description
Operating system	MS Windows operating systems family (MS Windows 11).
Database	PostgreSQL
Spring Boot	Spring Boot used for Server side Development V(3.2.5)
React.js	React for front-end development since it is an effective, declarative, and adaptable open-source JavaScript toolkit for creating straightforward, quick, and scalable frontend web application V(17.0.1).

Chapter 3.0.

Tools and Technologies Used

These are the necessary tools and materials needed to build the website both the frontend and the backend. These include software and open source materials. Technologies Used:

React JS

React is a declarative, efficient, and flexible JavaScript library for building user interfaces. It lets you compose complex UIs from small and isolated pieces of code called “components”.

React has a few different kinds of components, but we’ll start with **React**. Components, we use components to tell React what we want to see on the screen. When our data changes, React will efficiently update and re-render our components.

A component takes in parameters, called props (short for “properties”), and returns a hierarchy of views to display via the render method. The render method returns a *description* of what you want to see on the screen. React takes the description and displays the result. In particular, render returns a **React element**, which is a lightweight description of what to render. Most React developers use a special syntax called “**JSX**” which makes these structures easier to write.

React is a JavaScript-based UI development library. Facebook and an open-source developer community run it. Although React is a library rather than a language, it is widely used in web development. The library first appeared in *May 2013* and is now one of the most commonly used frontend libraries for web development. React offers various extensions for entire application architectural support.

Some of the important feature of the React are

JSX comes with the full power of JavaScript. You can put *any* JavaScript expressions within braces inside JSX. Each React element is a JavaScript object that you can store in a variable or pass around in your program. JSX is a JavaScript syntax extension. It isn't necessary to use JSX in React development, but it is recommended.

Components – React is all about components. You need to think of everything as a component. This will help you maintain the code when working on larger scale projects.

License – React is licensed under Facebook Inc. Documentation is licensed under CC BY4.0.

The Major React Advantages are

- Uses virtual DOM which is a JavaScript object. This will improve app's performance, since JavaScript virtual DOM is faster than the regular DOM.
- Can be used on client and server side as well as with other frameworks.
- Component and data patterns improve readability, which helps to maintain larger apps.

Java Spring Boot

Java Spring Framework (Spring Framework) is a popular, open source, enterprise-level framework for creating standalone, production-grade applications that run on the Java Virtual Machine (JVM). Java Spring Boot (Spring Boot) is a tool that makes developing web application and microservices with Spring Framework faster and easier through three core capabilities:

- Auto configuration
- An opinionated approach to configuration
- The ability to create standalone applications

Spring Boot is an open source Java-based framework used to create a Micro Service. It is developed by Pivotal Team. It is easy to create stand-alone and production ready spring applications using Spring Boot.

Spring Boot is a project that is built on the top of the Spring Framework. It provides an easier and faster way to set up, configure, and run both simple and web-based applications. It is a spring module that provides the RAD (Rapid Application Development) feature to the Spring Framework.

The Spring Framework (spring) is an open-source application framework that provides infrastructure support for developing Java applications. One of the most popular Java Enterprise Edition (Java EE) frameworks, spring helps developers create high performing applications using plain old Java objects (POJOs).

These features work together to provide you with a tool that allows you to set up a Spring-based application with minimal configuration and setup.

PostgreSQL

PostgreSQL is a powerful, open-source object-relational database management system (ORDBMS) with a strong reputation for reliability, feature robustness, and performance. Unlike simple relational databases, PostgreSQL extends the SQL language with features like complex queries, foreign keys, triggers, updatable views, and transactional integrity, making it highly extensible and compliant with ACID (Atomicity, Consistency, Isolation, Durability) properties. Known for its ability to handle heavy workloads and concurrent users efficiently, it is the preferred choice for enterprise-level applications—such as university management systems—where data accuracy, complex relationship mapping, and long-term scalability are critical requirements.

JAVASCRIPT

JavaScript is a high-level language which could be used independently or inculcated into the webpage. It can be used to handle requests and responses and also add dynamic behavior and also store information on a website.

JavaScript (often shortened to JS) is a lightweight, interpreted, object-oriented language with first-class functions, and is best known as the scripting language for Web pages, but it's used in many non-browser environments as well.

JavaScript is a scripting language that enables you to dynamically update content, control multimedia, animate images, and pretty much everything else. (Okay, not everything, but it is amazing what you can achieve with a few lines of JavaScript code.)

JavaScript is used by programmers across the world to create dynamic and interactive web content like applications and browsers. JavaScript is so popular that it's the most used programming language in the world, used as a client-side programming language by 97.0% of all websites.

JAVA EE

Java EE is a structured application with a separate client, business, and Enterprise layers. It is mostly used to develop APIs for Desktop Applications like antivirus software, game, etc. It is mainly used for developing web applications. Suitable for beginning Java developers.

A java platform is a distinct software environment for running applications written in java. The four java platforms are Java SE, Java EE, Java ME and JavaFx.

The Java EE is a superset of the Java SE, and it provides a runtime environment, technologies and API for building and running enterprise web applications. All API in java SE are also contained in Java EE.

Java Platform Enterprise Edition, An application software platform from Oracle based on the Java programming language. Originally developed by Sun, which Oracle acquired in 2010, Java EE services are performed in the middle tier between the user's machine and the enterprise's databases and legacy information systems. Java EE comprises a specification, reference implementation and set of testing suites. Its core component is

Enterprise JavaBeans (EJBs), followed by Java Server Pages (JSPs) and Java servlets and a variety of interfaces for linking to the information resources in the enterprise. The Java EE interfaces include JDBC for databases, JNDI for directories, JTA for transactions, JMS for messaging, Java Mail for email systems and Java IDL for CORBA connectivity. Java Connectors are interfaces to a variety of legacy applications.

Spring Tool Soot(STS Editor)

Eclipse STS validates your project and provides quick fixes for your applications. For example, when working with Spring Data JPA, the IDE may be used to validate query method names (more on this in section 6).

STS also provides a graphical view on all bean methods and their mutual relationships. You may want to have a closer look at the graphical editors that come with STS by looking into the views that are available under the menus *window*, *show view* and then *Spring* respectively.

STS also offers other additional useful features that are not limited to Spring applications only.

Visual Studio Code

Visual Studio Code is a source-code editor made by Microsoft for Windows, Linux and macOS.

Visual Studio Code is a streamlined code editor with support for development operations like debugging, task running, and version control. It aims to provide just the tools a developer needs for a quick code-build-debug cycle and leaves more complex workflows to fuller featured IDEs, such as Visual Studio IDE.

Features include support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git. Users can change the theme, keyboard shortcuts, preferences, and install extensions that add additional functionality.

In the Stack Overflow 2021 Developer Survey, Visual Studio Code was ranked the most popular developer environment tool, with 70% of 82,000 respondents reporting that they use it.

Postman

Postman is an API (application programming interface) development tool which helps to build, test and modify APIs. Almost any functionality that could be needed by any developer is encapsulated in this tool. It is used by over 5 million developers every month to make their API development easy and simple.

Postman is an application used for API testing. It is an HTTP client that tests HTTP requests, utilizing a graphical user interface, through which we obtain different types of responses that need to be subsequently validated.

An HTTP request method includes Request Method; Request URL, Request Headers, Request Body, Pre-request Script, and Tests. Let's talk about these Request methods one by one... Request Methods: You will find several types of Request methods in POSTMAN. Depending on your requirements or test you can choose one of them.

Chapter 4.0.

Application and Description

1) Sequence Diagram

This diagram traces the exact flow of data when a new student registers, showing how your React Frontend talks to the Spring Boot Backend and PostgreSQL Database.

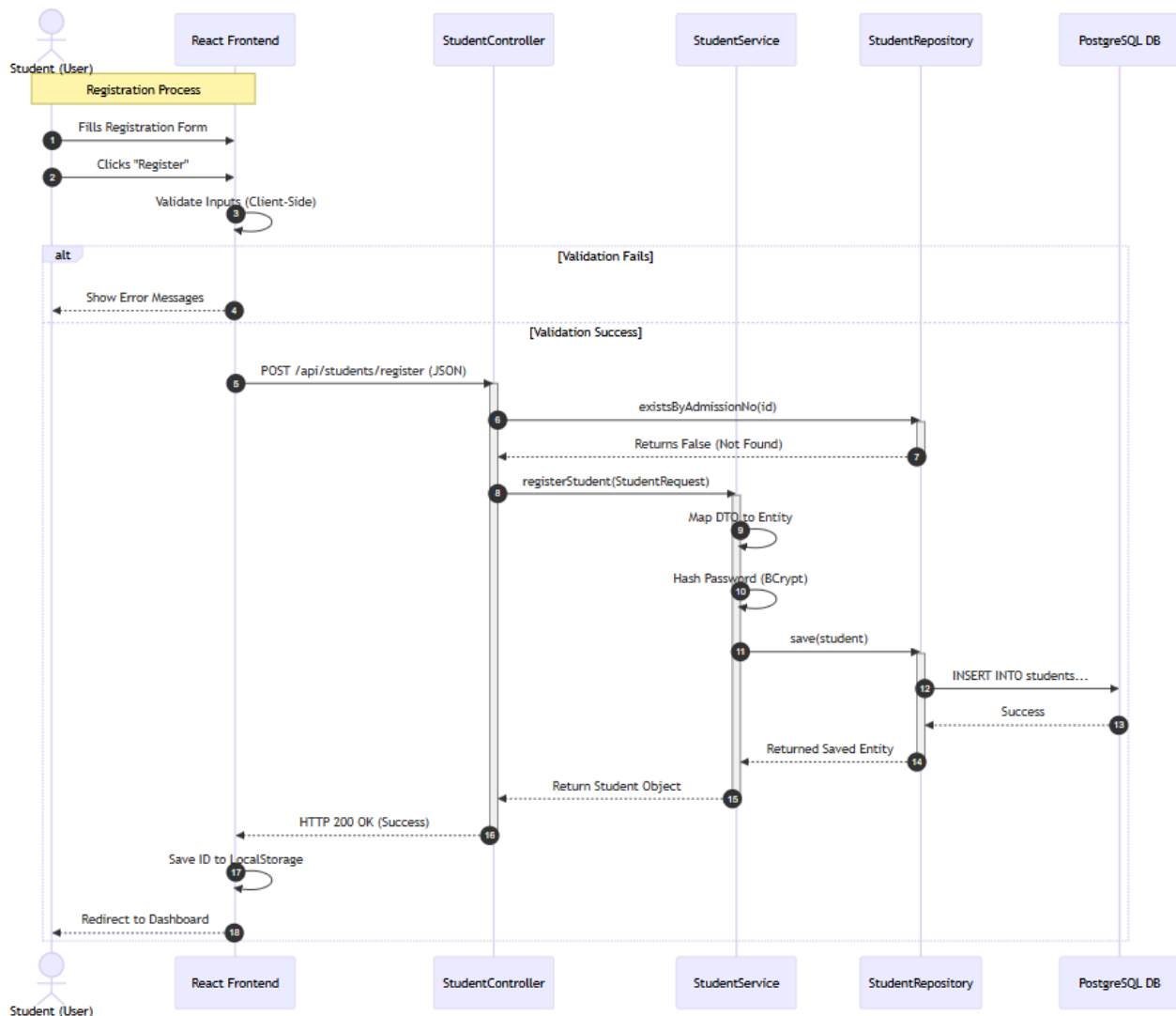


Figure 1 Sequence Diagram (Student Registration)

2) Sequence Diagram (View Exam Schedule)

This diagram illustrates the logic we fixed earlier—fetching a schedule based on the student's specific Branch and Semester.

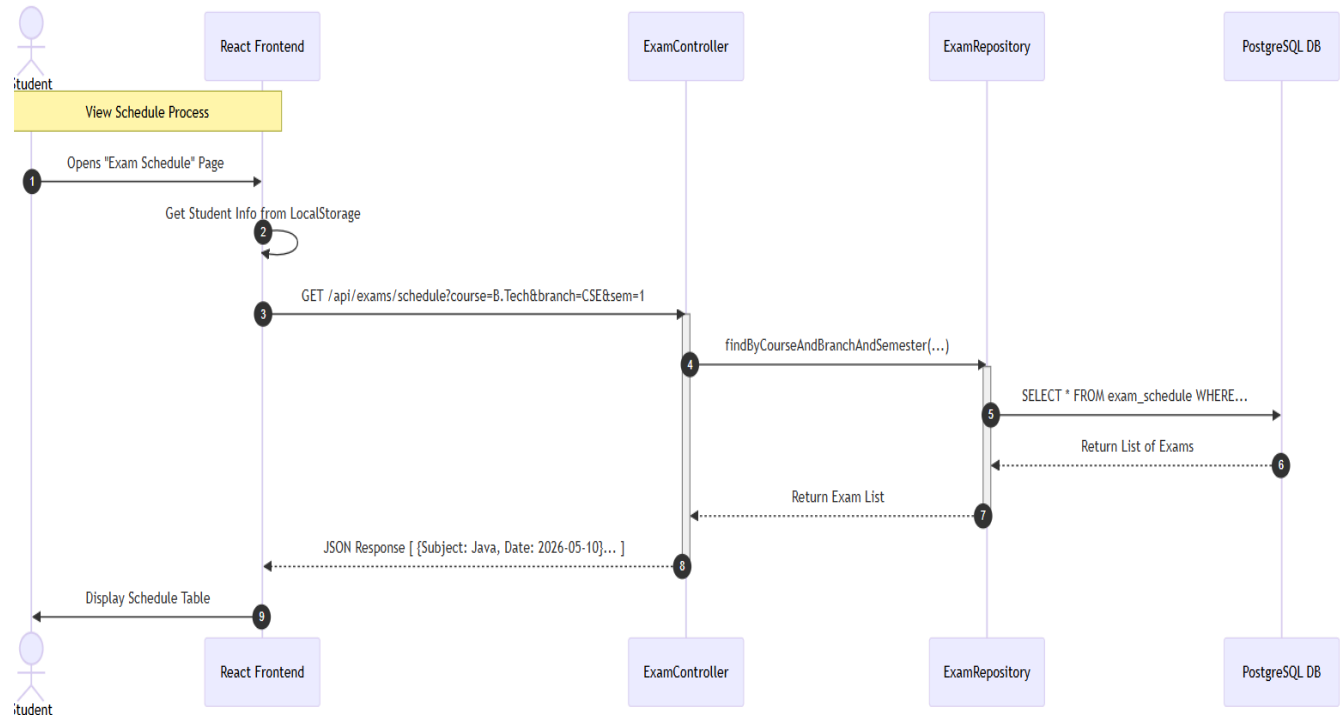


Figure 2 Sequence Diagram (View Exam Schedule)

2) Class Diagram

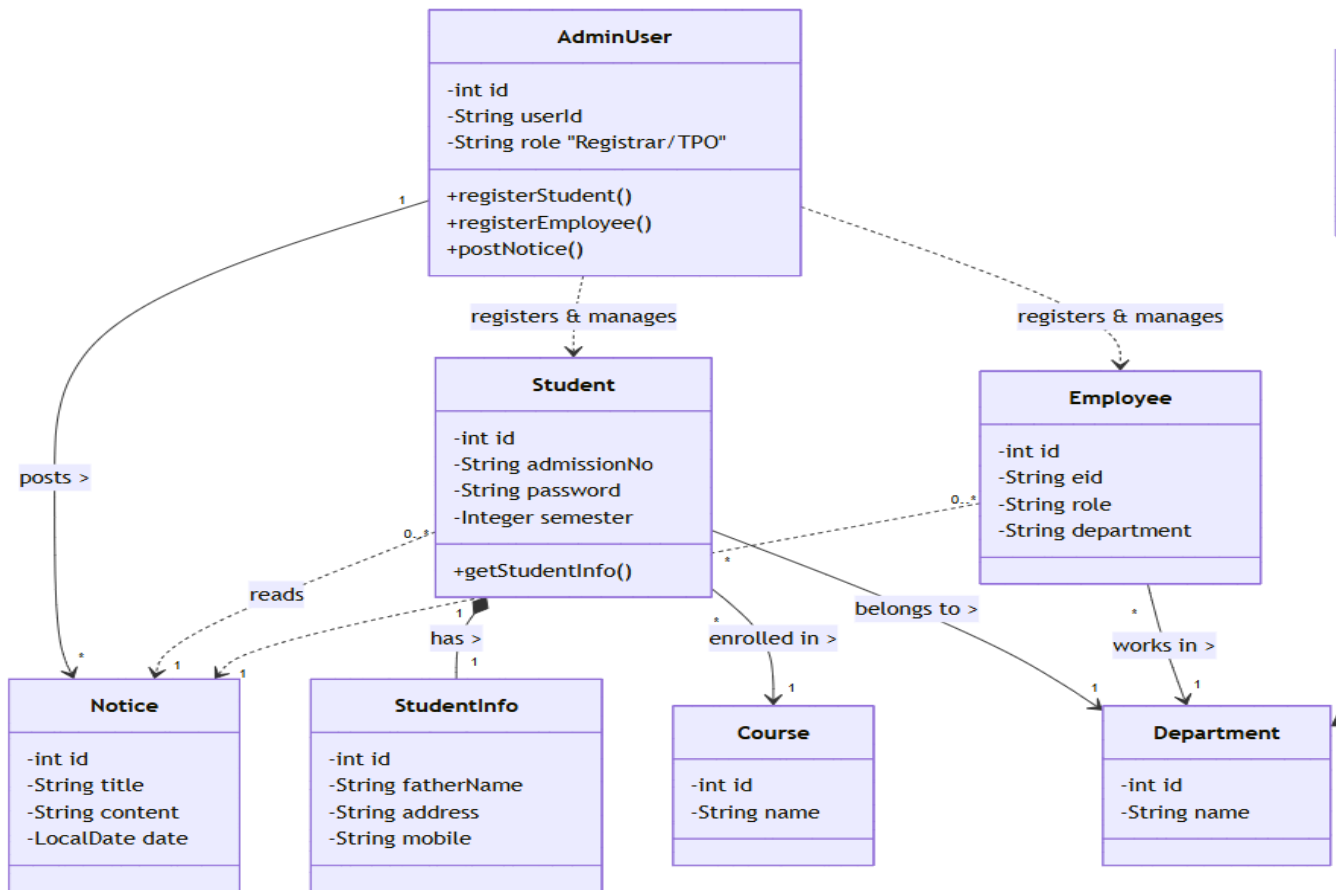


Figure 3 Class Diagram

3) Use Case Diagram

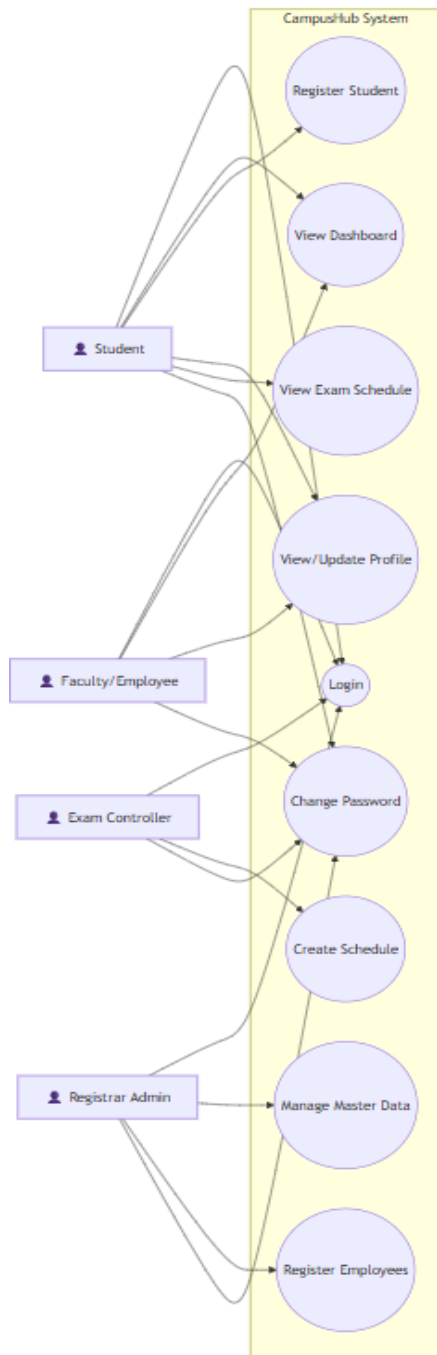
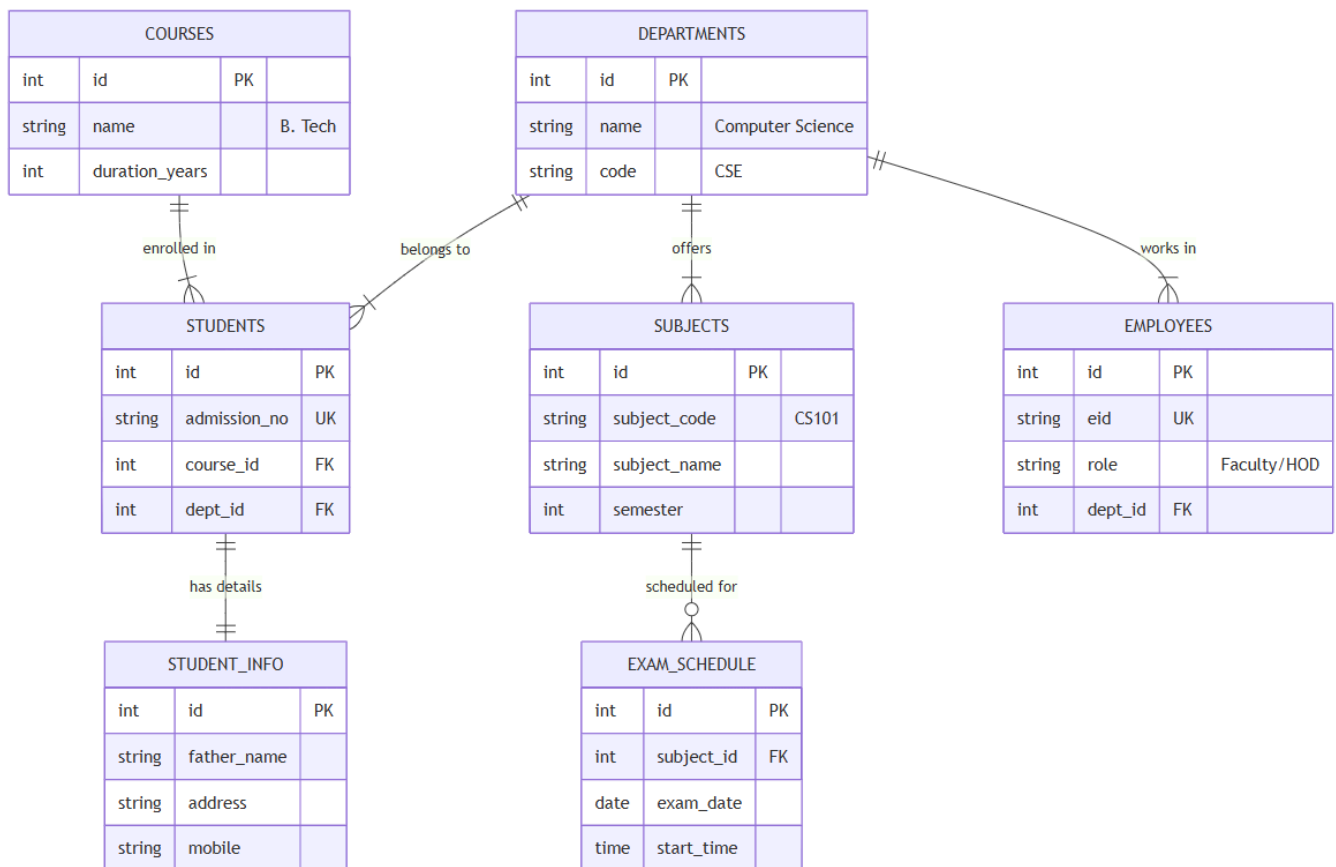


Figure 3 Use Case Diagram

Database Design and GUI

The database design for CampusHub is built on a **relational model using PostgreSQL**, structured to ensure data integrity, scalability, and performance. At its core, the system separates **Authentication Data** from **Profile Data**; the primary **students** and **employees** tables store essential login credentials and academic identifiers (like Admission Number and Password) for fast access, while detailed personal information (such as addresses and family details) is offloaded to linked **student_info** and **employee_info** tables. To avoid redundancy and hardcoding, the system utilizes **Master Data tables** for **courses**, **departments** (branches), and **subjects**, which are dynamically linked to users via foreign keys. Additionally, operational tables like **exam_schedule** are logically mapped to these academic structures, ensuring that schedules are automatically associated with the correct set of students based on their registered branch and semester.



Chapter 6.0

Implementation

Description

This application is based entirely on free source applications and with the intention to cut costs. All of the functionalities need to be carried out in an orderly manner. The technologies used for the project are ReactJS, Java EE and PostgreSQL.

Implementation of functions

User Interface and admin interface are the main interfaces created in this project. With the User and admin interface, we used ReactJS. The JSX was used to create the structure of the page while the bootstrap is used in styling of pages. JavaScript is dynamic language, so we can make use of asynchronous API calls. These API calls are used for transferring data from front-end to back-end and vice versa.

Rest API

Restful API is used in the backend. All the data that is passed between front-end and back-end is in JSON format and uses Http Protocol. The REST API is secured with Spring security and JWT. The response with appropriate status code is sent wrapped in the Response Entity to the frontend. Example of JSON response is provided below:

Basic React Code

ReactJS is used as a view layer as it is one of the most popular programming frameworks of UI. The requests are sent and received using the 'axios' API of React. The JWT tokens in front end are managed with the help of local storage. The tokens are stored in local storage of the browser and are used in further requests until they are expired. At the time of logout, the tokens are removed from the local storage.

Login Module

Admin and user login are provided separately. Users and admin can use their logging into the application. Spring security will validate the username and password from the database and after validation the appropriate access and refresh tokens are sent to the user. The access token is valid for 5 minutes and the refresh token is valid for 20 minutes. These can be used as authorization tokens for further requests.

Chapter 7.0

Project Snapshot

1) Login/Home Page

Welcome to CampusHub

Center for Development of Advanced Computing, Noida

Role:

User ID:

Password:

Login

[Register New Student](#)

[Register New Employee](#)

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2) Registrar/Admin Dashboard

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Logout

Hello, registrar!

[Add Course](#)

[Add Department](#)

[Add Subject](#)

[View Employees](#)

[View Students](#)

[View Feedback](#)

3) Add Course Page

Add Courses

Course Name:

-- Select Course --

Duration (Years):

Add Course

Existing Courses

B. SC (3 Years)

B. TECH (4 Years)

BCA (3 Years)

M. SC (2 Years)

M. TECH (2 Years)

MCA (2 Years)

Back

4) Add Department

Add Departments

Department Name:

-- Select Department --



Dept Code:

Add Department

Existing Departments

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (Code: AIML101)

BIOTECHNOLOGY (Code: BIO102)

CHEMISTRY (Code: CHEM103)

CIVIL ENGINEERING (Code: CE104)

CLOUD COMPUTING (Code: CC105)

COMPUTER APPLICATIONS (Code: CA106)

COMPUTER SCIENCE AND ENGINEERING (Code: CSE107)

DATA SCIENCE (Code: DS108)

ELECTRICAL ENGINEERING (Code: EE109)

ELECTRONICS AND COMMUNICATION ENGINEERING (Code: ECE110)

INFORMATION TECHNOLOGY (Code: IT111)

MATHEMATICS (Code: MATH112)

MECHANICAL ENGINEERING (Code: ME113)

MOBILE APPLICATION DEVELOPMENT (Code: MAD114)

PHYSICS (Code: PHY115)

5) View Employee

Employee Management

[← Back to Dashboard](#)

Filter By Department: -- All Departments -- [Refresh List](#)

Emp ID	Name	Role	Department	Mobile	Action	
654	Raghav Devgan	examController	COMPUTER SCIENCE AND ENGINEERING	6325417890	Modify	Delete
101	Gaurav Joshi	faculty	DATA SCIENCE	8794662130	Modify	Delete
102	Neeraj Kumar	faculty	COMPUTER SCIENCE AND ENGINEERING	9874563215	Modify	Delete
201	Dheeraj Jha	hod	DATA SCIENCE	7788994455	Modify	Delete

6) View Feedback

System Feedback

[Back](#)

STUDENT

25/01/2026

hello this is a sample feedback

EMPLOYEE

25/01/2026

hello this is a sample feedback from faculty

EMPLOYEE

25/01/2026

hello this is a sample feedback 3

EMPLOYEE

25/01/2026

hello this is a sample feedback from exam controller

STUDENT

28/01/2026

Hello this is a feedback

EMPLOYEE

28/01/2026

nice infrastructure!

STUDENT

29/01/2026

werwtr

STUDENT

29/01/2026

sgdhasgd

6) Student Dashboard

Center for Development of Advanced Computing, Noida

Logout

Hello, Shikhar Sharma!
Admission No: 546
Course: B. TECH
Branch: DATA SCIENCE

[Profile](#)

[Notice Board](#)

[Feedback](#)

[Change Password](#)

7) Faculty/HOD/ExamDashboard

Center for Development of Advanced Computing, Noida

Logout

Hello, Gaurav Joshi!
Employee ID: 101
Department: Data Science

[Profile](#)

[Notice](#)

[Feedback](#)

[Change Password](#)

8) Profile Page

Employee Profile

Official Details

Name:	Gaurav Joshi	Employee ID:	101
Role:	faculty	Department:	Data Science
Email:	gaurav@gmail.com	Mobile:	8794662130

Personal & Professional Details

Qualification:	Postgraduation	Experience:	7 years
Date of Birth:	1986-02-20	Address:	Not Updated
PAN Card:	QWERT1234E		

Update Profile

Back

9) Student Registration

Center for Development of Advanced Computing, Noida

Student Registration

*Session:	-- select session --
*Course:	-- select course --
*Branch:	-- select branch --
*Semester:	-- select semester --
*Section:	-- select section --
*Admission No:	
*Student Name:	
*Mobile No:	
*Email:	

9) Registration Form (With Client-Side Validation)

Employee Registration

*Type:	Faculty
*Course:	B. TECH
*Department:	ARTIFICIAL INTELLIGENCE & MACHINE LEARNI
*Employee ID:	
	ID is required
*Employee Name:	
	Employee Name is required
*Mobile No:	
	Mobile Number is required
*Email:	
	Email is required
Address:	
*Password:	..
	Password must be 8+ chars, with Upper, Lower, Number, and Special Char
*Confirm Password:	
	Passwords do not match

Chapter 8.0

Conclusion

CampusHub, the College Management System, has been designed and developed to provide an efficient, reliable, and user-friendly platform for managing academic and administrative activities within an educational institution. The system integrates essential modules such as administration, faculty management, and student services into a single centralized solution, thereby reducing manual effort and improving overall operational efficiency.

By automating processes such as student admissions, course management, attendance tracking, examination handling, and result generation, CampusHub ensures accuracy, transparency, and timely access to information for all users. The system facilitates smooth interaction between administrators, faculty members, and students, enabling better communication and streamlined workflows.

CampusHub also offers scalability and flexibility, allowing institutions to adapt the system to their specific requirements and future growth. Overall, the implementation of CampusHub contributes to enhanced productivity, better data management, and improved decision-making, making it a valuable solution for modern educational institutions.

Chapter 9.0

References

React Documentation

- <https://react.dev/>

Spring Boot Documentation

- <https://docs.spring.io/spring-boot/index.html>

Spring Security Reference

- <https://docs.spring.io/spring-security/reference/index.html>

Axios HTTP Client

- <https://axios-http.com/docs/intro>

PostgreSQL Documentation

- <https://www.postgresql.org/docs/current/>

Java Documentation (Oracle)

- <https://docs.oracle.com/en/java/>

Spring Tools Suite (STS)

- <https://spring.io/tools>